

PAUL MITCHELL

MACHINE LEARNING ENGINEERING | ARTIFICIAL INTELLIGENCE & DATA SYSTEMS

Riverdale, MD | (413) 204-0493 | pjmittell@wm.edu | linkedin.com/in/paul-mitchell-b1497b30b | github.com/pjmittell7

PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML concentration) and B.S. graduate specializing in deep learning, NLP transformers, predictive modeling, and real-time AI security systems. Hands-on experience developing end-to-end cloud data pipelines on AWS, fine-tuning transformer models, and deploying sub-millisecond evaluation proxies (0.038ms P99). Proven track record translating complex unstructured data and adversarial attack patterns into high-accuracy, production-ready machine learning solutions.

EDUCATION

College of William & Mary

Williamsburg, VA

Master of Science in Computer Science (AI/ML Concentration) | Expected Dec 2026 | GPA: 3.95

Bachelor of Science in Computer Science | Graduated Dec 2024

Relevant Coursework: Machine Learning Systems, Deep Learning, Data Mining, Advanced Software Engineering

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, scikit-learn, LightGBM, DistilBERT, Sentence-BERT, BERTopic, Hugging Face, NLP transformers

Data Engineering & Cloud: Python (Pandas, NumPy, spaCy), SQL (PostgreSQL), AWS (S3, EC2, Lambda), DVC, Docker, Git/GitLab CI

Analytics & Observability: Streamlit, Grafana, Prometheus, statistical modeling, feature engineering, latency profiling

Software & Systems: FastAPI, Java, C/C++, Bash, Linux (Ubuntu), REST APIs, test-driven development, asynchronous programming

TECHNICAL & PROFESSIONAL EXPERIENCE

Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote

Jun 2025 - Sep 2025

- Built an automated ETL pipeline with Python, spaCy, and AWS S3 to clean and structure ~20,000 multi-topic records, cutting data preparation cycles from hours to minutes.
- Fine-tuned DistilBERT for text classification ($F1 = 0.89$) and applied BERTopic clustering to isolate recurring operational patterns from unstructured text corpora.
- Developed a LightGBM predictive classification model and deployed an interactive Streamlit dashboard delivering real-time decision metrics for program leadership.
- Engineered structured data extraction workflows querying survey databases to generate on-demand operational summaries for program mentors.
- Implemented cloud dataset versioning using DVC on AWS S3 to ensure reproducible data lineages across multi-cohort model evaluations.

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA

Feb 2025 - Present

- Architected telemetry pipelines using Prometheus and Grafana, capturing real-time hardware metrics across departmental compute clusters to eliminate training bottlenecks.
- Administer and support 100+ Linux instructional compute servers, maintaining system integrity, static routing, and resource management across multi-user environments.
- Automated repetitive systems auditing and user permission validation using Bash and Puppet, ensuring consistent environment configurations.
- Conducted architectural evaluations of multi-user Docker containerization, documenting GPU pass-through isolation boundaries and resource overhead.

KEY TECHNICAL & SYSTEMS PROJECTS

Real-Time Adversarial Prompt Defense Proxy | Python, FastAPI, Docker

2026

- Engineered an asynchronous reverse proxy evaluating incoming HTTP streams in memory at 0.038ms P99 latency, neutralizing steganographic injection and payload assembly attacks prior to model ingestion.
- Built sub-millisecond classification pipelines maintaining false positive rates under 0.1% across benchmark evaluation suites.

Code Analysis & Condition Prediction Engine | Python, Hugging Face

2025

- Engineered an automated data pipeline harvesting 7,677 open-source repositories and fine-tuned CodeT5 on 50,000 masked Python methods to predict conditional logic statements.
- Achieved BLEU score of 75.98 and 33.6% exact match accuracy, evaluating token representations across syntactically complex code structures.

Automotive CAN Bus Threat Classification Pipeline | Python, PyTorch

2025

- Built an end-to-end ML classification pipeline parsing raw automotive CAN network traffic to detect payload injection, replay attacks, and denial-of-service anomalies.
- Trained ensemble classifiers isolating anomalous flow signatures with 99.1% detection precision under high-throughput conditions.